

PH 600: Research Methodology

Course Code	PH 600
Course Name	Research Methodology
L-T-P-C	1-0-0-1
Intended for	MS/Ph.D.
Mutual Exclusion	RM-600 in other schools
Approval	54th BoA (May 2024)

Course Contents

- **Module I:** Formal communication of email and verbal, Teacher-student relationship, Technical and scientific presentation Skills (3 Hours)
- **Module II:** Ethics in research, Copyright and plagiarism, Interdisciplinary research work, Project management (3 Hours)
- **Module III:** Safety and precautions, best practices in experimental as well as simulation research (2 Hours)
- **Module IV:** Literature Survey, Designing of the experimental and theoretical research, Data analysis and presentation (3 Hours)
- **Module V:** Drafting the reports, Reading and Writing research paper (3 Hours)

Laboratory Modules

1. A technical presentation on the specific topic and write a report on the same topic in journal format.
2. Task on data analysis of scientific data sets.
3. Group discussions on a specific topic.

Assessment will be based on a combination of individual and group assignments, presentations, and a final research report.

Text Books

1. Michael P. Marder, **Research Methods for Science**, Cambridge University Press, 2012.
2. Caroline Van Den Brul, Crackle And Fizz, **Essential Communication And Pitching Skills For Scientists**, Imperial College Press , 2014.

References

1. E.M. Phillips and D.S. Pugh, **How to get a Ph.D.?**, UBSPD, 1993.
2. Wyne C. Booth, Colomb, William, **The Craft of Research**, University of Chicago Press, eBook.
3. Ranjit Kumar, **Research Methodology: A Step-by-Step Guide for Beginners**, Sage South Asia, 2011.
4. R Panneerselvam, **Research Methodology**, PHI Learning, 2009.
5. C. R. Kothari, **Research Methodology: Methods and Techniques**, New Age International, 2004.